Backbone-conditioned sidechain reconstruction improves prioritization of de novo protein binders

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Abstract

De novo protein design generates thousands of candidate complexes, yet most fail experimentally and co-folding every candidate is expensive. Because design protocols pack sidechains inside the complex, supplied coordinates can mask steric and polar incompatibility relative to the monomer backbone. Here we introduce `BC-Score`, which strips supplied sidechains, predicts monomer-conditioned rotamer distributions with an SE(3)-equivariant graph neural network, and averages 38 within-target and 29 cross-target descriptors without fitted weights. Across 3,532 tested binders (13 targets), standalone `BC-Score` reaches 0.383 Average Precision versus 0.135 for `PyRosetta` ΔG_{sep}, 0.120 for `SpatialPPI v2` ($P < 0.001$), and 0.361 for `AlphaFold3 ipSAE` ($P = 0.338$) at 0.043 s per complex on an A100 GPU. A 25% `BC-Score`-to-`AlphaFold3` cascade recovers 66 versus 67 unfiltered top-10-per-target hits, while hybrids reach 0.449–0.508 Average Precision and improve `Nipah-G` top-50 and `Scorpion` top-20 recovery despite equal or lower `Scorpion` top-50 recovery and lower standalone precision on unrelaxed coordinates.

1 Introduction

De novo protein binder design has advanced through generative backbone modeling [1, 2], structure-conditioned inverse folding [3, 4, 5, 6], all-atom co-folding networks [7, 8, 9, 10, 11, 12, 13, 14], and integrated design pipelines [15, 16]. Yet retrospective analyses show that most computationally generated candidates fail to bind in vitro [17, 18, 19]. In the Overath et al. benchmark [19], only 391 of 3,532 single-chain receptor designs (11.07%) bind experimentally. Force-field energies such as `PyRosetta` ΔG_{sep} [20, 21, 22] and graph classifiers such as `SpatialPPI v2` [23] offer limited ranking precision (pooled global Average Precision [AP] = 0.135 and 0.120), whereas co-folding all 3,532 complexes with `Boltz-1` [11] or `AlphaFold3` [10] requires 34–177 NVIDIA A100 GPU-hours.

A central limitation of scoring supplied design coordinates directly is that upstream protocols (`ProteinMPNN` [4] with `Rosetta FastRelax` [24], rotamer packing [25, 26, 27, 28, 29], or joint diffusion) optimize sidechain torsions within the fixed complex pose $\mathbf{x}_0$. Supplied sidechains $\mathbf{s}_i^{\text{supplied}}$ thus reflect partner-conditioned packing rather than the intrinsic preference of the isolated monomer backbone [30], potentially masking strained polar rotamers [31] and sterically forced aromatics.

Here we introduce the **Backbone-Conditioned Sidechain Reconstruction Score (`BC-Score`)**, a fast scoring function that strips non-backbone atoms from each chain and applies a three-layer SE(3)-equivariant GNN [32, 6] pretrained on crystal monomers [33, 34] (Fig. 1a). From monomer backbone geometry ($\text{N}, \text{C}_\alpha, \text{C}, \text{O}, \text{C}_\beta$) and sequence alone, the network predicts each residue's expected sidechain centroid $\hat{\mathbf{s}}_i^{\text{recon}}$, distal functional tip $\hat{\mathbf{t}}_i^{\text{recon}}$, and a $K = 3$ Mixture Density Network (MDN) [35] over χ_1 rotamer modes $\{\pi_{i,k}, \hat{\mathbf{s}}_{i,k}^{\text{recon}}, \hat{\mathbf{t}}_{i,k}^{\text{recon}}\}_{k=1}^{3}$ [26]. Comparing $\hat{\mathbf{s}}_i^{\text{recon}}$ with $\mathbf{s}_i^{\text{supplied}}$ yields a model-derived proxy for packing inconsistency—the **sidechain reconstruction discrepancy** $\|\hat{\mathbf{s}}_i^{\text{recon}} - \mathbf{s}_i^{\text{supplied}}\|$ and minimum-mode rotamer distance $\min_k \|\hat{\mathbf{s}}_{i,k}^{\text{recon}} - \mathbf{s}_i^{\text{supplied}}\|$ (Fig. 1b,c)—while providing partner-blind sidechain coordinates for interface scoring.

Default `BC-Score` combines 38 within-target descriptors (weighted equally at 1/38) and 29 cross-target descriptors (weighted equally at 1/29, with zero feature weights fitted on benchmark targets) organized into five physical modules: (C1) monomer foldability, 3D inverse-folding compatibility, and Spatial Aggregation Propensity (`SAP`) [36]; (C2) sidechain reconstruction discrepancy, rotamer strain proxies, and autograd backbone restoring forces ($\nabla_{\mathbf{X}} \log P_{\text{IF}}(\mathbf{s} \mid \mathcal{G})$); (C3) bilateral buried unsatisfied polar ("Buns") penalties [31], burial-screened electrostatics, and aromatic O-ring packing [37]; (C4) in-place alchemical sequence counterfactuals, Alanine scanning, and holo-versus-apo inverse-folding log-odds; and (C5) continuous six-degree-of-freedom (6-DoF) rigid-body relaxation and 24-pose SE(3) re-docking funnel sharpness under a pretrained pairwise interface potential. Unless explicitly labeled otherwise, `BC-Score` refers to this default descriptor-level equal-weight model (38 within-target and 29 cross-target descriptors); as sensitivity analyses, we also report a five-module macro-average variant that averages the five module sub-scores at 1/5 each and includes an additional C4 cross-target inverse-folding term (30 cross-target descriptors total; `BC-Score-5M`), inner-Leave-One-Target-Out (LOTO) shrinkage-regularized descriptor weights on the default 38-within-target and 29-cross-target descriptor set (`BC-Score [Reg. LOTO]`), and a cohort-free single-complex ($M = 1$) thermodynamic-

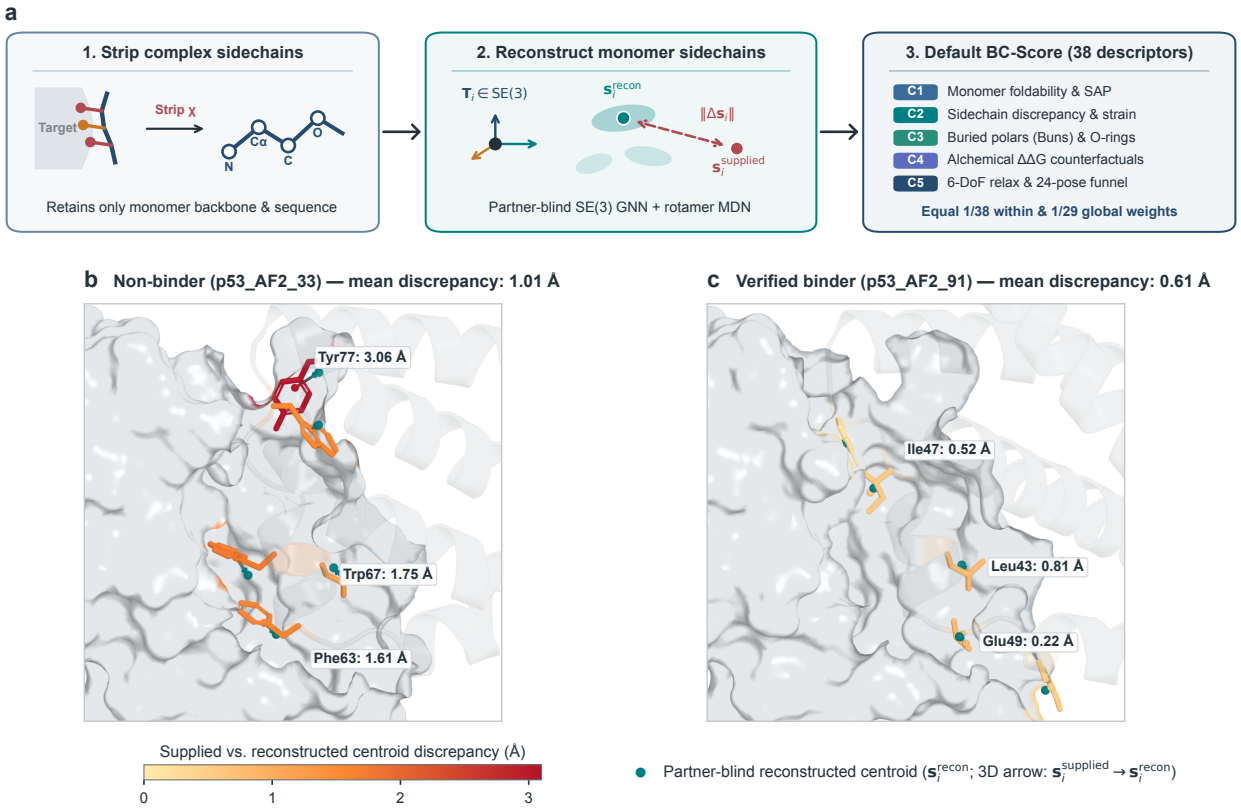


Figure 1. BC-Score pipeline and model-predicted sidechain reconstruction discrepancy at designed Mdm2 interfaces. **a**, Three-stage `BC-Score` evaluation: (1) supplied sidechains are stripped from the input complex PDB structure, retaining only monomer backbone atoms (N, C_α, C, O) and sequence; (2) a pretrained partner-free SE(3)-equivariant GNN and $K = 3$ rotamer Mixture Density Network (MDN) predict monomer-conditioned sidechain centroids $\mathbf{s}_i^{\text{recon}}$, functional tips $\mathbf{t}_i^{\text{recon}}$, and model-derived packing discrepancy $\|\Delta\mathbf{s}_i\| = \|\mathbf{s}_i^{\text{recon}} - \mathbf{s}_i^{\text{supplied}}\|$; and (3) default `BC-Score` combines 38 within-target and 29 cross-target descriptors across five physical modules (C1–C5) using equal 1/38 within-target and 1/29 cross-target descriptor weights without benchmark-fitted feature weights (0.043 s per complex in batched A100 GPU evaluation; the five-module macro-average variant `BC-Score-5M` averages the five module sub-scores at 1/5 each and includes a fourth C4 cross-target inverse-folding term [30 cross-target descriptors total]). **b, c**, Interface close-ups of an experimentally tested Mdm2 non-binder (`p53_AF2_33`, **b**; $n = 1$ complex) and verified binder (`p53_AF2_91`, **c**; $n = 1$ complex) from the Overath et al. benchmark ($n = 3,532$ complexes; two-sided Mann–Whitney U test on interface strain Z_{strain}, $P = 1.1 \times 10^{-4}$). Interface residues are colored by centroid reconstruction discrepancy on a shared 0–3.1 Å blue-to-red colour scale (1.01 Å full-interface mean for the non-binder versus 0.61 Å for the binder), with 3D displacement arrows connecting complex-packed supplied centroids ($\mathbf{s}_i^{\text{supplied}}$) to partner-blind reconstructed centroids ($\mathbf{s}_i^{\text{recon}}$, filled teal spheres).

cycle formulation (`True-EBM`).

2 Results

2.1 Backbone-conditioned sidechain reconstruction

On 60 held-out crystal validation monomers (12,248 residues) and 50 crystal complexes, our 2-Point SE(3) GNN reduces sidechain centroid Mean Absolute Error (MAE) from 0.89 Å to 0.48 Å (86.0% versus 65.6% within 1.0 Å) and distal tip MAE from 1.59 Å to 0.86 Å relative to a canonical rigid-backbone local-frame template (Extended Data Table 1). The largest gain occurs on bulky aromatics (Phe, Trp, Tyr), where centroid MAE drops from 2.13 Å to 0.58 Å and tip MAE drops from 3.81 Å to 0.90 Å; at crystal complex interfaces (< 8.5 Å C_α), centroid MAE is 0.57 Å versus 1.05 Å for the template.

Across the Overath et al. benchmark ($n = 3,532$; 391 binders, 3,141 non-binders; Fig. 1b,c), model-predicted discrepancy between supplied and reconstructed sidechains is higher at interface residues than at non-interface surface residues on the same monomer (two-sided paired Wilcoxon signed-rank test, $P = 1.4 \times 10^{-13}$). Between outcomes, experimental non-binders exhibit larger within-target interface discrepancy ($P = 1.5 \times 10^{-3}$) and a higher uncertainty-normalized strain proxy ($Z_{\text{strain}} = +0.14$ versus -0.07, two-sided Mann–Whitney U test, $P = 1.1 \times 10^{-4}$) than verified binders.

Because $\hat{\mathbf{s}}_i^{\text{recon}}$ is predicted from static crystal monomer backbones rather than experimental unbound structures, we interpret Z_{strain} as a statistical packing-inconsistency signal rather than a direct physical measurement of unbound conformational strain.

2.2 Standalone benchmark across 13 targets

We evaluated standalone `BC-Score` (operating strictly on input design PDB structures without co-folding) against `PyRosetta` ΔG_{sep} [21, 22], `SpatialPPI v2` [23], and seven co-folding metrics from Overath et al. [19] across all 3,532 complexes and 13 targets under LOTO cross-validation (Table 1, Fig. 2, and Extended Data Table 2). Co-folding baselines include `AlphaFold2` initial-guess interface predicted Aligned Error (`iPAE`) [18], `ColabFold AF2-Multimer` [8, 9] interface predicted Structural Aligned Error (`ipSAE`) [38] and active interface predicted Template Modeling score (`actifpTM`) [39], `Boltz-1 ipSAE` [11, 38], and `AlphaFold3` [10] interface predicted Template Modeling score (`ipTM`), Local Interaction Score (`LIS`) [40], and `ipSAE`.

Across all 3,532 complexes, standalone `BC-Score` nearly triples the Pooled Global AP of `PyRosetta` ΔG_{sep} and `SpatialPPI v2` (0.383 versus 0.135 and 0.120, $\Delta = +0.248$, $P < 0.001$, Benjamini–Hochberg false discovery rate [FDR] $q = 0.001$; 13-target Mean AP = 0.559 versus 0.267 and 0.174; Table 1 and Fig. 2a). Across formulation and weighting variants, shrinkage-regularized inner-LOTO descriptor weights on the identical 38-within-target and 29-cross-target descriptor set yield Pooled AP = 0.382 ($\Delta = -0.001$ versus default equal descriptor weights, $P = 0.786$; 0.560 Mean Target AP), while the five-module macro-average variant (`BC-Score-5M`, which both weights the five module subscores at 1/5 each and includes a fourth C4 cross-target term) reaches Pooled AP = 0.366 (0.527 Mean Target AP; Extended Data Table 2).

Compared against GPU co-folders, standalone `BC-Score` exceeds `AlphaFold2-Initial-Guess iPAE`, `ColabFold AF2-Multimer ipSAE` and `actifpTM`, `AlphaFold3 ipTM`, and `Boltz-1 ipSAE` on Pooled Global AP (0.383 versus 0.242–0.336, all $P \leq 0.036$, BH $q \leq 0.046$) and 13-target Mean AP (0.559 versus 0.376–0.480). Against `AlphaFold3 LIS` (0.350, $P = 0.126$) and `AlphaFold3 ipSAE` (Pooled AP = 0.361, $\Delta = +0.022$, 95% confidence interval [CI] $[-0.023, +0.064]$, $P = 0.338$; 13-target Mean AP = 0.545, $\Delta = +0.014$, $P = 0.802$), the observed differences are not statistically significant across the 13 targets, which indicates comparable precision on this cohort without establishing formal statistical equivalence. Excluding `FGFR2` (2,123 of 3,532 designs) widens the pooled difference between `BC-Score` and both `Boltz-1 ipSAE` and `AlphaFold3 ipSAE` (0.487 versus 0.353 and 0.390). Across the 11 multi-hit campaigns (≥ 6 binders), `BC-Score` achieves a Mean AP of 0.502 versus 0.430 for `Boltz-1 ipSAE`, 0.496 for `AlphaFold3 LIS`, and 0.543 for `AlphaFold3 ipSAE` (Fig. 2b,c).

Performance holds on receptors non-homologous to the pretraining corpus (Smith–Waterman local identity $< 30\%$; Section 4.1): across the 10 targets with $< 30\%$ identity to the interface training set ($n = 2,933$), `BC-Score` achieves Pooled AP = 0.401 and Mean Target AP = 0.590 (versus 0.367 and 0.536 for `AlphaFold3 ipSAE`), and across the 5 targets with $< 30\%$ identity to both monomer and interface corpora ($n = 2,326$), Mean Target AP is 0.643 (versus 0.582 for `AlphaFold3 ipSAE` and 0.206 for `PyRosetta` ΔG_{sep}).

2.3 Sidechain attribution and component dissection

To isolate the contribution of reconstructed sidechains across all 3,532 complexes (Fig. 3a), we compared a monomer-backbone-only baseline (`C1` alone; Pooled AP = 0.199, 11-target Mean AP = 0.226) against three matched five-module (`C1`–`C5`) evaluations that hold standardization, the 24-pose SE(3) re-docking grid, and the five alchemical counterfactuals fixed while varying only the sidechain coordinate source: canonical rigid-template sidechains $\hat{\mathbf{s}}_i^{\text{template}}$ (0.265, 0.344), complex-packed supplied sidechains $\mathbf{s}_i^{\text{supplied}}$ (0.281, 0.381), and partner-blind GNN-reconstructed sidechains $\hat{\mathbf{s}}_i^{\text{recon}}$ in `BC-Score` (0.383, 0.502), which improves Pooled Global AP by +0.102 and 11-target Mean AP by +0.121 over supplied sidechains (both $P < 0.001$).

Across the 11 multi-hit campaigns, different modules lead on different targets relative to positive prevalence (0.065–0.573; Fig. 3b): sidechain discrepancy and rotamer strain (C2) lead on five targets (`Mdm2`, `TrkA`, `Pdl1`, `IL7Ra`, `InsulinR`), polar satisfaction (C3) leads on polar interfaces (`IL2Ra`, `sntx`), monomer foldability (C1) leads on `FGFR2`, alchemical specificity (C4) leads on `SARS-CoV-2 RBD`, and pose funnel sharpness (C5) leads on `EGFR` and `VirB8`.

2.4 Two-stage screening cascades and hybrid reranking

We evaluated `BC-Score` as a Stage-1 pre-filter before GPU co-folding and as a Stage-2 hybrid score across retention budgets $\rho \in [0\%, 100\%]$ (Fig. 4, Extended Data Fig. 1a–c, and Extended Data Table 3).

At **Operating Point 1 ($\rho = 0\%$, 0.042 A100 GPU-hours)**, standalone `BC-Score` without co-folding recovers 56 top-10-per-target and 111 global top-200 binders, exceeding unfiltered `AlphaFold3 ipSAE` on global top-200 recovery (89 hits) as well as `PyRosetta` ΔG_{sep} (25) and `SpatialPPI v2` (32; Fig. 4b and Extended Data Table 3).

At **Operating Point 2 ($\rho = 25\%$, 44.4 A100 GPU-hours)**, Stage-1 `BC-Score` retains 52.2% of all 391

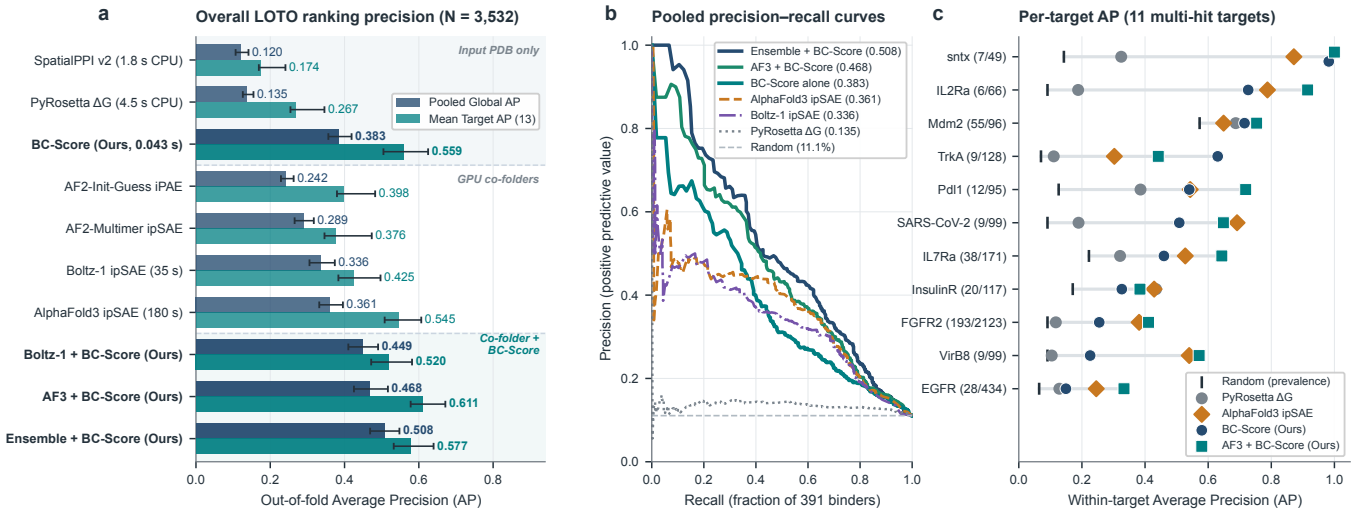


Figure 2. Leave-One-Target-Out (LOTO) ranking performance across 3,532 experimentally tested de novo binders. **a**, Out-of-fold Pooled Global AP (filled navy bars, point estimate) and unweighted 13-target Mean Target AP (filled teal bars, mean across 13 targets) with error bars indicating 95% target-stratified paired bootstrap confidence intervals ($B = 10,000$ resamples across $n = 3,532$ complexes; two-sided paired bootstrap $P < 0.001$ for standalone `BC-Score` versus `PyRosetta` ΔG_{sep}, `SpatialPPI v2`, `AF2-Init-Guess iPAE`, and `AF2-Multimer ipSAE`; $P = 0.036$ versus `Boltz-1 ipSAE`; $P = 0.338$ versus `AlphaFold3 ipSAE`; Table 1 and Extended Data Table 2). The vertical grey dashed line marks random baseline prevalence (0.111). **b**, Pooled out-of-fold precision–recall curves across all $n = 3,532$ candidates (391 verified binders; horizontal grey dashed line indicates random prevalence 11.1%). **c**, Within-target out-of-fold AP across the 11 multi-hit receptor campaigns (≥ 6 binders per target, $n = 3,477$ of 3,532 designs), with positive counts (n_{pos}/N_t) indicated in each y-axis label and target-specific random prevalence marked by vertical black tick marks.

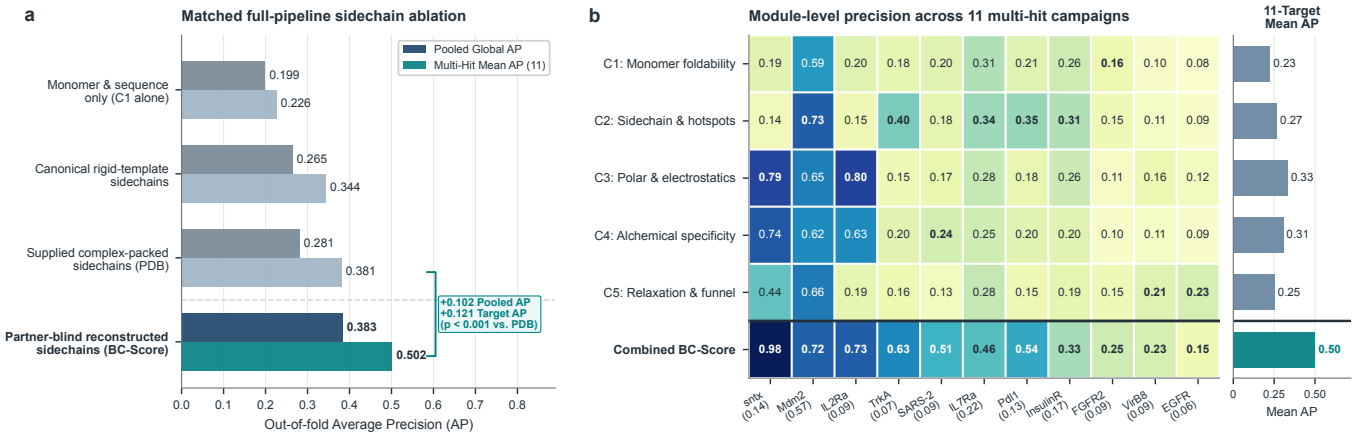


Figure 3. Sidechain attribution ablation and module-level precision across 11 multi-hit campaigns. **a**, Out-of-fold Pooled Global AP (filled navy bars, left) and 11-target Multi-Hit Mean AP (filled teal bars, right) with error bars showing 95% target-stratified paired bootstrap confidence intervals ($B = 10,000$ resamples across $n = 3,532$ complexes; two-sided paired bootstrap $P < 0.001$ for partner-blind GNN-reconstructed sidechains $\hat{\mathbf{s}}_i^{\text{recon}}$ versus supplied sidechains $\mathbf{s}_i^{\text{supplied}}$, rigid-backbone template $\hat{\mathbf{s}}_i^{\text{template}}$, and monomer-only `C1`). **b**, Heat map of within-target AP for each physical module (C1–C5) and combined `BC-Score` across the 11 multi-hit campaigns ($n = 3,477$ complexes; positive prevalence in parentheses), with right-hand horizontal bars showing the unweighted 11-target Mean AP.

binders globally (47.8% discarded; 64.3% mean across targets) versus 35.8% for `PyRosetta` ΔG_{sep} and 21.2% for `SpatialPPI v2` (Extended Data Fig. 1c). Despite this 52.2% overall recall cutoff, ranking the 25% survivors with `AlphaFold3 ipSAE` alone yields 66 top-10-per-target hits (versus 67 from unfiltered `AlphaFold3`: 45 of the original 67 top-10 hits survive Stage 1 while 22 are filtered out, and 21 other surviving true binders move into the top 10 after high-`ipSAE` non-binders are removed) and 113 global top-200 hits (69 and 119 with the `AlphaFold3 + BC-Score` hybrid, Pooled AP = 0.414). At $\rho = 50\%$ (88.5 GPU-hours, 76.7% recall), `AlphaFold3 ipSAE` alone recovers

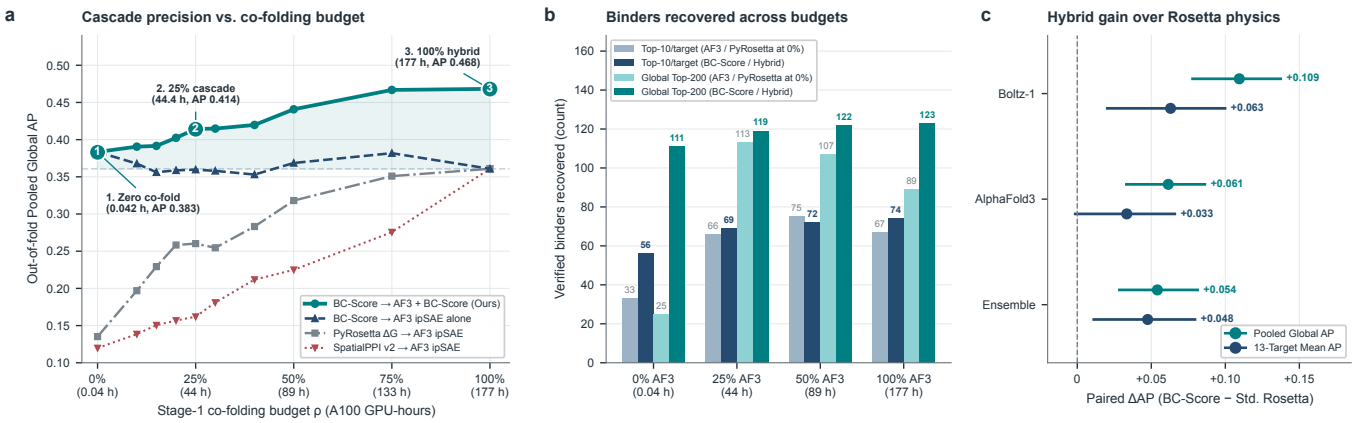


Figure 4. Two-stage screening cascades and post-co-fold hybrid reranking across 3,532 candidates. **a**, Out-of-fold Pooled Global AP across Stage-1 co-folding retention budgets $\rho \in [0\%, 100\%]$ (0.042–176.6 A100 GPU-hours across $n = 3,532$ complexes), comparing `BC-Score` (solid teal and navy curves), `PyRosetta` ΔG_{sep} (orange dashed curve), and `SpatialPPI v2` (red dotted curve) pre-filters followed by `AlphaFold3 ipSAE` alone or the `AlphaFold3 + BC-Score` hybrid (numbered circles denote Operating Points **1**: 0% co-folding; **2**: 25% cascade; **3**: 100% hybrid). **b**, Verified binders recovered in the Top-10 per target (13 targets, maximum 130) and Global Top-200 ($n = 3,532$) at $\rho \in \{0\%, 25\%, 50\%, 100\%\}$ (colored bars indicate method and budget). **c**, Point estimates (filled circles) and horizontal error bars showing 95% target-stratified paired bootstrap confidence intervals ($B = 10,000$ resamples across $n = 3,532$ complexes) for Paired ΔAP (Pooled Global AP in teal, 13-Target Mean AP in navy; all two-sided paired bootstrap $P < 0.001$) when augmenting `Boltz-1`, `AlphaFold3`, or their ensemble with `BC-Score` instead of standard Rosetta physical descriptors (vertical grey dashed line marks ΔAP = 0).

more top-10-per-target hits than the hybrid (75 versus 72), whereas the hybrid recovers more global top-200 hits (122 versus 107; Extended Data Table 3).

At **Operating Point 3 (ρ = 100%, 176.6 A100 GPU-hours)**, combining `BC-Score` with co-folder confidence under LOTO cross-validation raises Pooled Global AP to 0.449 with `Boltz-1`, 0.468 with `AlphaFold3` (13-target Mean AP = 0.611), and 0.508 with their ensemble (all $P < 0.001$; Table 1), outperforming standard Rosetta descriptors (`SAP` [36], `CamSol` [41], ΔG_{sep}, and ΔG_{sep} per buried Solvent-Accessible Surface Area [$\Delta G_{\text{sep}}/\Delta$SASA]; ΔAP = +0.054–0.109; Fig. 4c and Extended Data Fig. 1a,b).

2.5 Same-epitope discrimination and in silico partition-function co-design

Across same-target binder and non-binder pairs sharing ≥ 50% (326,474 pairs) to ≥ 90% (4,124 pairs) Jaccard contact overlap (C_α < 8.5 Å), `BC-Score` ranks the verified binder higher in 67.2%–73.9% of pairs (66.5%–72.7% for `BC-Score-5M`), beating `PyRosetta` ΔG_{sep} (59.9%–64.3%), supplied-sidechain scoring (58.0%–60.0%), `ESM-2` [42] (54.9%–56.6%), and backbone inverse folding (49.9%–52.1%; Extended Data Fig. 2).

To test whether the potential can guide computational sequence redesign on fixed backbones (39 failed Overath scaffolds, 3 per target; Extended Data Figs. 1d–f and 3), we compared single-pose energy minimization $E(\mathbf{x}_0, \mathbf{s})$ (∼ 0.12 s per complex) against partition-function free-energy minimization. Optimizing $E(\mathbf{x}_0, \mathbf{s})$ alone improves the single-pose score on 35 of 39 scaffolds (two-sided Wilcoxon signed-rank test, $P = 1.3 \times 10^{-9}$) and Buns on 31 of 39 ($P = 3.7 \times 10^{-6}$), but degrades 24-pose funnel Root-Mean-Square Deviation (RMSD) on 19 of 39 scaffolds (20 of 39 improved, $P = 0.72$) via apolar mutations that also stabilize decoy poses (Extended Data Fig. 3a). Minimizing the partition-function free energy $\Delta F_{\text{pose}}(\mathbf{x}_0, \mathbf{s}) = E_\theta(\mathbf{x}_0, \mathbf{s}) + \tau \log\left(\frac{1}{M_{\text{dec}}} \sum_{m \in \text{Decoy}} e^{-E_\theta(\mathbf{x}_m, \mathbf{s})/\tau}\right)$ over the 24-pose SE(3) ensemble resolves this in silico mode: it lowers interface energy on 34 of 39 scaffolds ($P = 2.1 \times 10^{-8}$), reduces Buns on 33 of 39 ($P = 8.9 \times 10^{-8}$), increases Alanine-scan $\Delta\Delta G$ on 34 of 39 (+1.01 → +2.10, $P = 2.1 \times 10^{-8}$), improves frozen monomer inverse-folding log-likelihood on 38 of 39 ($P = 7.3 \times 10^{-12}$), sharpens the 24-pose funnel on **37** of **39** scaffolds (0.95 Å → 0.74 Å, $P = 2.3 \times 10^{-9}$; Extended Data Fig. 3b), and reduces pose-funnel RMSD by 52.4% (2.22 Å → 1.06 Å) on an independent held-out 24-pose SE(3) verification ensemble across **33** of **39** scaffolds ($P = 1.3 \times 10^{-4}$). These results demonstrate computational objective consistency across held-out poses, not experimental wet-lab rescue.

2.6 Held-out external validation

Using the frozen Overath checkpoints and normalization parameters (Section 4.6), we evaluated two held-out public campaigns (Fig. 5 and Extended Data Table 4): (1) `Nipah-G`, from the Adaptyv Bio and Proteinbase *Nipah Competition Results* collection (`nipah-binder-competition-results`,

https://proteinbase.com/collections/nipah-binder-competition-results; n = 881 expressed candidates with Boltz-2 complex predictions [12] after excluding 146 non-expressed, 2 structure-missing, and 1 conflicting entry from the 1,030 evaluated submissions, yielding 100 surface plasmon resonance [SPR] binders [11.35%]); and (2) Scorpion, from the Anthropic and Adaptyv Bio scorpion toxin binder design dataset (Anthropic/scorpion-toxin-binder-design, https://huggingface.co/datasets/Anthropic/scorpion-toxin-binder-design; n = 243 raw pre-co-fold design PDB structures across 5 neurotoxins [receptor target structures from PDB 25IH, 1NPI, 1AHO, 7K18 and AlphaFold Database AF-P01479-F1] generated using Mosaic, PXDesign, Genie 3, RFdiffusion3 and RFdiffusion [1], Proteina-Complexa, FreeBindCraft [15], BoltzGen [16], Protein Hunter, and BoltzDesign1; 21 Adaptyv SPR hits [8.64%] and 12 Twist-confirmed publication hits [4.94%]).

External performance is metric- and variant-dependent (Extended Data Table 4). On Nipah-G (0.041 s per complex in batched A100 evaluation), standalone default BC-Score achieves AP = 0.155 and 10/50 top-50 hits (0.156 and 11/50 with regularized LOTO weights; M = 1 True-EBM reaches AP = 0.182, 11/50 top-50 hits), outperforming ProteinMPNN [4], ESMFold monomer predicted Local Distance Difference Test (pLDDT) [42], and Boltz-2 pDockQ2 [43, 44], LIS [40], ipTM, and complex pLDDT (AP = 0.100–0.146), while trailing standalone Boltz-2 ipSAE (AP = 0.193, Area Under the Receiver Operating Characteristic curve [AUROC] = 0.622, 7/25 top-25, 9/50 top-50) and min ipSAE (AP = 0.229, AUROC = 0.645, 9/25 top-25, 10/50 top-50; Fig. 5a). Combining default BC-Score with Boltz-2 ipSAE raises AP by +18.7% (0.193 → 0.229) and top-50 hits from 9/50 (18%) to **16/50** (32.0%), tying min ipSAE in AP while trailing min ipSAE in AUROC (0.618 versus 0.645) and top-25 hits (8/25 versus 9/25).

On Scorpion (n = 243 raw pre-co-fold PDB structures across 5 targets), standalone default BC-Score (Pooled AP = 0.141, AUROC = 0.618, 3/20 top-20, 7/50 top-50) and BC-Score-5M (Pooled AP = 0.170, AUROC = 0.609, 3/20 top-20, 7/50 top-50) trail standalone co-folders (Protenix-V2 ipSAE [13] Pooled AP = 0.226, 5/20 top-20, 11/50 top-50; ESMFold2 ipSAE [45] Pooled AP = 0.336, 8/20 top-20, 16/50 top-50; self-consistency DockQ [scDockQ] [46] baselines are shown in Extended Data Table 4). One possible explanation for this lower pooled standalone precision is that the supplied Scorpion coordinates combine unrelaxed pre-co-fold structures across ten upstream generators in small per-target cohorts (N_t = 38–66), which may shift cross-target descriptor baselines relative to uniformly relaxed Overath complexes, although differences in target difficulty, small-toxin receptor geometry, and assay definitions may also contribute. Within targets, default BC-Score reaches 0.319 5-target Mean AP (0.448 on the 12 publication hits) and BC-Score-5M reaches 0.408 (0.599 on publication hits, driven by ranking both LqhIII hits at positions 1–2; across the four targets with ≥ 3 hits, BC-Score and BC-Score-5M achieve 0.316 and 0.261 versus 0.281 for Protenix-V2 and 0.347 for ESMFold2; Fig. 5b). Post-co-fold hybrids improve early precision and within-target AP for both co-folders—Protenix-V2 + BC-Score lifts Pooled AP (0.226 → 0.288), 5-target Mean AP (0.262 → 0.367; 0.224 → 0.415 on publication hits), and top-20 hits (5/20 → 8/20), while ESMFold2 + BC-Score lifts Pooled AP (0.336 → 0.376), 5-target Mean AP (0.377 → 0.463; 0.386 → 0.527 on publication hits), and top-20 hits (8/20 → 10/20)—but exhibit lower global AUROC (0.706 versus 0.748 and 0.785 versus 0.823) and equal or lower top-50 recovery (11/50 versus 11/50 for Protenix-V2, and 13/50 versus 16/50 for ESMFold2).

3 Discussion

Reconstructing monomer-conditioned sidechain geometry from backbone coordinates provides an inexpensive post-design structural signal (0.038–0.043 s per complex in batched A100 GPU evaluation, 4,176–4,783× faster than AlphaFold3 at the post-design scoring step, excluding upstream backbone and sequence generation) that complements all-atom co-folding networks. At the same time, three limitations bound its scope and guide practical deployment: (1) **Recall versus compute in Stage-1 prefiltering:** while a ρ = 25% Stage-1 cutoff recovers 66 versus 67 AlphaFold3 top-10-per-target hits (45 shared plus 21 newly promoted after filtering out high-ipSAE non-binders), it discards 47.8% of all binders globally (52.2% recall); campaigns prioritizing overall recall should use ρ = 50% (76.7% recall; Extended Data Table 3) or post-co-fold hybrid reranking (ρ = 100%). (2) **Reduced standalone transfer on heterogeneous external cohorts:** on Scorpion, pooled standalone BC-Score trails co-folding baselines, potentially reflecting baseline shifts across unrelaxed coordinates from multiple upstream generators, smaller per-target sample sizes, or target-specific effects; in such settings, BC-Score is more effective as a post-co-fold hybrid modifier or via BC-Score-5M for within-target ranking. (3) **Model-derived proxy and lack of prospective wet-lab validation:** because our GNN is trained on static crystal monomers rather than solution ensembles, sidechain discrepancy is a statistical proxy rather than a direct physical measurement of unbound strain, and all co-design trajectories represent *in silico* objective optimization pending prospective experimental synthesis.

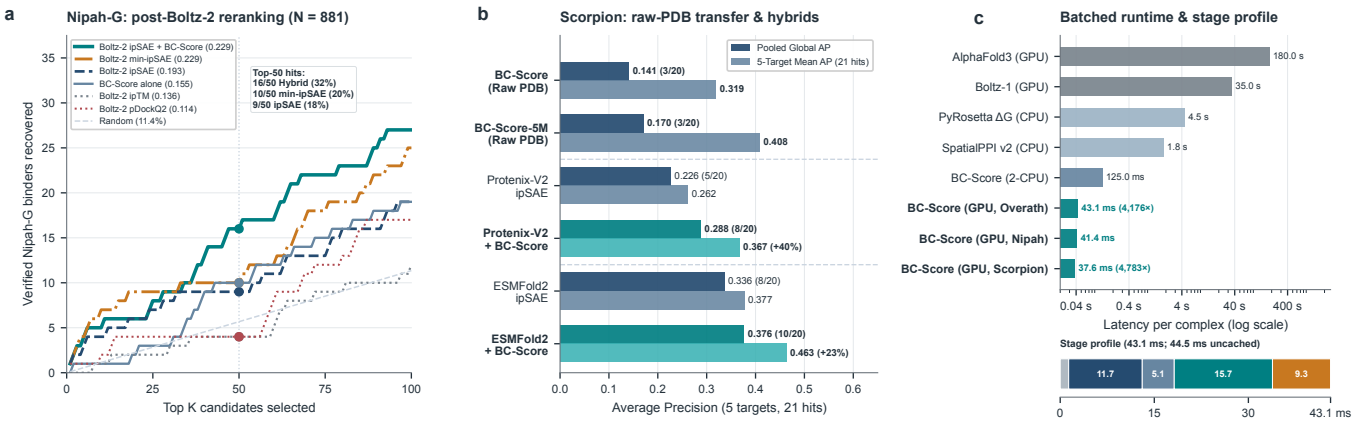


Figure 5. Held-out external validation across two public campaigns and batched runtime profiling. a, Cumulative verified binders recovered in the top $K \in [1, 100]$ selections on the Adaptyv Bio `Nipah-G` campaign ($n = 881$ `Boltz-2` complexes, 100 binders; grey dotted line marks random expectation), comparing standalone default `BC-Score` (solid blue curve), `Boltz-2` confidence metrics (navy dashed, orange dash-dotted, grey dotted, and red dotted curves), and the locked `Boltz-2 ipSAE + BC-Score` hybrid (solid teal curve; 95% bootstrap confidence intervals [$B = 10,000$ resamples] for Pooled AP are reported in Extended Data Table 4). **b**, Point estimates for Pooled Global AP (dark navy and dark teal bars, with Top-20 hit counts in parentheses) and unweighted 5-target Mean Target AP (light blue and light teal bars) across $n = 243$ raw pre-co-fold `Scorpion` neurotoxin design PDB structures (5 targets, 21 Adaptyv hits; sensitivity metrics on the 12 Twist-confirmed publication hits are reported in brackets in Extended Data Table 4). **c**, Mean batched per-complex latency (log scale, horizontal bars) across GPU co-folders ($n = 3,532$), CPU baselines ($n = 100$), and `BC-Score` ($n = 100$ complexes on Overath, $n = 881$ on `Nipah-G`, $n = 243$ on `Scorpion`), with the A100 GPU stage profile below (41.75 ms across five GPU compute passes plus 1.35 ms cached-target coordinate setup and transfer = 43.10 ms batched wall-clock; replacing cached setup with full uncached raw-PDB parsing of both chains [2.70 ms] gives $41.75 + 2.70 = 44.45$ ms).

4 Methods

4.1 Pretraining corpus and sequence-identity audit

All three neural networks within `BC-Score`—the 3D backbone inverse-folding GNN, the 2-Point sidechain reconstruction GNN, and the pairwise interface potential—were pretrained on crystal structures prior to evaluation on de novo design benchmarks. Combining 900 crystal structures (50–450 residues) from `DiscoTope` [33] with non-redundant crystal chains from `SAbDab` [34] clustered via `MMseqs2` [47] (240 clusters) yielded 1,288 monomer chains, split deterministically (seed 2026) into 1,228 training and 60 validation monomers. To form 520 native-packed training interfaces without de novo designed binders, we combined 360 cleaved contiguous crystal subdomain interfaces (40–58 residues packing against the remaining core with $\geq$ 18 C_α contacts within 8.5 Å and a $\geq$ 6-residue cleavage buffer) with 160 non-redundant `SAbDab` interfaces.

We quantified sequence overlap with the Overath et al. benchmark (3,532 complexes, 13 targets) via Smith–Waterman local alignment (match = +1, mismatch = −1, gap open = −2, gap extend = −0.5, normalized by shorter sequence length). De novo binder sequences have a median maximum local identity of 14.1% to any pretraining chain (Interquartile Range [IQR] 12.8%–15.8%, 95th percentile 19.4%, maximum 30.0%). Among the 13 target receptors, 10 targets (`FGFR2`, `IL10Ra`, `IL7Ra`, `InsulinR`, `LTK`, `Mdm2`, `Pdl1`, `TrkA`, `VirB8`, `sntx`; 2,933 of 3,532 complexes [83.0%]) share < 30% local identity (11.5%–23.8%) with the interface training corpus, while 3 targets (`EGFR`, `IL2Ra`, `SARS-CoV-2 RBD`) have crystal homologues in `SAbDab` and `DiscoTope`. Against the combined monomer and interface corpora, 5 targets (`FGFR2` [15.9%], `IL10Ra` [17.3%], `LTK` [19.2%], `VirB8` [11.5%], `sntx` [18.3%]; $n = 2,326$ [65.9%]) have < 30% local identity to every pretraining chain.

4.2 Network architectures and training hyperparameters

For each residue i, we construct a local frame $\mathbf{T}_i = (\mathbf{R}_i, \mathbf{x}_{i,C_\alpha}) \in \mathrm{SE}(3)$ from $(\mathrm{N}, \mathrm{C}_\alpha, \mathrm{C})$ via Gram–Schmidt orthogonalization and connect $k = 16$ spatial nearest neighbors with 41 SE(3)-invariant edge features and 57 node features [32, 6]. All networks were trained in PyTorch using `AdamW` (weight decay 10^{-4}, gradient clipping at 1.0, cosine schedule to $\eta_{\min} = 10^{-4}$, seed 2026):

1. **3D Backbone Inverse-Folding GNN:** 3 layers, hidden dimension 96, dropout 0.10, trained on backbone coordinates $(\mathrm{N}, \mathrm{C}_\alpha, \mathrm{C}, \mathrm{O}$, ideal $\mathrm{C}_\beta)$ for 15 epochs (batch size 8, initial learning rate 1.5×10^{-3}, cross-entropy loss with label smoothing 0.05).

2. **2-Point Sidechain Reconstruction GNN:** 3 layers, hidden dimension 96, conditioned on backbone

geometry and one-hot residue identity s_i. It predicts local-frame centroid and tip offsets and uncertainties $(\hat{\mathbf{u}}_i^{\mathrm{cm}}, \hat{\sigma}_i^{\mathrm{cm}}, \hat{\mathbf{u}}_i^{\mathrm{tip}}, \hat{\sigma}_i^{\mathrm{tip}})$ plus a $K = 3$ rotamer MDN head $\{\pi_{i,k}, \hat{\mathbf{u}}_{i,k}^{\mathrm{cm}}, \hat{\sigma}_{i,k}^{\mathrm{cm}}, \hat{\mathbf{u}}_{i,k}^{\mathrm{tip}}, \hat{\sigma}_{i,k}^{\mathrm{tip}}\}_{k=1}^{3}$ [35, 26], trained for 15 epochs (batch size 8, learning rate 2.5×10^{-3}) on $\mathcal{L}_{\mathrm{SC}} = \mathcal{L}_{\text{cm-NLL}} + \|\hat{\mathbf{u}}_i^{\mathrm{cm}} - \mathbf{u}_i^{\mathrm{cm}}\|_2^2 + 0.35(\mathcal{L}_{\text{MDN-NLL}} + \mathcal{L}_{\text{MDN-WTA}}) + 0.85(\mathcal{L}_{\text{tip-NLL}} + \|\hat{\mathbf{u}}_i^{\mathrm{tip}} - \mathbf{u}_i^{\mathrm{tip}}\|_2^2 + 0.35[1 - \cos\angle(\hat{\mathbf{u}}_i^{\mathrm{tip}}, \mathbf{u}_i^{\mathrm{tip}})])$.

3. **Pairwise Interface Potential:** 2 monomer GNN layers (hidden dimension 64) and a 3-layer pairwise multilayer perceptron ($173 \to 64 \to 32 \to 2$) over cross-interface pairs within 10.5 Å C_α, using variance-convolved steric widths $\sigma_{\mathrm{eff},ij}^2 = \sigma_0^2 + (\hat{\sigma}_i^{\mathrm{cm}})^2 + (\hat{\sigma}_j^{\mathrm{cm}})^2$ and burial-screened dielectric $\varepsilon(B_{ij})$. Trained for 12 epochs (1 interface with 24 rigid-body poses and 5 sequence decoys per batch, learning rate 2.0×10^{-3}) using InfoNCE ($\tau = 0.50$) plus hinge margins $0.5 \max(0, E_0 - E_{\mathrm{pose}}^{\min} + 0.50) + 0.6 \max(0, E_0 - E_{\mathrm{seq}}^{\min} + 0.35)$ for decoys with RMSD > 2.0 Å.

4.3 Physical scoring architecture and calibration sensitivity

Default `BC-Score` organizes 38 within-target descriptors $\mathbf{z}_i^{\mathrm{within}} \in \mathbb{R}^{38}$ $(9 + 9 + 10 + 6 + 4)$ and 29 cross-target descriptors $\mathbf{z}_i^{\mathrm{global}} \in \mathbb{R}^{29}$ $(5 + 10 + 5 + 3 + 6)$ into five modules:

1. **C1 (Monomer foldability; 9 within-target, 5 cross-target):** Intra-monomer backbone H-bonds, buried unsatisfied backbone polars, `SAP` [36], steric clash, interface α-helix fraction and length, β-sheet and loop penalties, and weakest-link 5-residue sliding-window 3D inverse-folding log-likelihood $(\min_m \frac{1}{5} \sum_{j=m}^{m+4} \log P_{\mathrm{IF}}(s_j \mid \mathcal{G}_{\mathrm{apo}}))$.

2. **C2 (Sidechain discrepancy, strain & backbone forces; 9 within-target, 10 cross-target):** Hotspot sidechain Z-strain, minimum-mode rotamer distance $(\min_k \|\hat{\mathbf{s}}_{i,k}^{\mathrm{recon}} - \mathbf{s}_i^{\mathrm{supplied}}\|)$, interface-minus-surface discrepancy, template discrepancy contrast, reconstructed-versus-supplied clash jump, buried rotamer entropy cost, aromatic O-ring packing [37], holo re-packing gain, and autograd backbone restoring forces $(\mathbf{F}_i^{\mathrm{bb}} = \nabla_{\mathbf{X}_i} \log P_{\mathrm{IF}}(\mathbf{s} \mid \mathcal{G}))$.

3. **C3 (Polar satisfaction & electrostatics; 10 within-target, 5 cross-target):** Bilateral buried unsatisfied polar ("Buns") penalties [31] on binder and receptor tips, burial-weighted H-bond and salt-bridge networks, Coulombic electrostatics on native and surface-decoy poses, soft van der Waals packing, and partner-blind interface energy per contact.

4. **C4 (Alchemical specificity & holo-vs-apo inverse folding; 6 within-target, 3 cross-target [4 in `BC-Score-5M`]):** Five fixed-backbone sequence counterfactuals—(i–ii) two interface permutations, (iii) deep-core polar injection ($C_\alpha < 6.8$ Å to D, E, K, R, N, Q), (iv) computational Alanine scanning ($C_\alpha < 9.0$ Å to Ala), and (v) core–rim O-ring and charge inversion—plus holo-minus-apo inverse-folding log-odds and `ESM-2 8M` [42] hydrophobic frustration (in `BC-Score-5M`, the C4 cross-target sub-score additionally includes the complex-minus-monomer 3D inverse-folding log-odds sum, giving 30 cross-target terms across the five module averages).

5. **C5 (6-DoF relaxation & 24-pose funnel; 4 within-target, 6 cross-target):** Boltzmann-weighted C_α RMSD and Z-score across a 24-pose SE(3) ensemble (input $\mathbf{x}_0$, 12 local normal and tangent shifts and ±12° tilts, 11 surface ray-cast decoys), minimum energy under continuous 6-DoF Langevin relaxation on $(\boldsymbol{\omega}, \mathbf{t}) \in \mathfrak{se}(3)$, shear stiffness, epitope concavity, and cohort contact consensus.

To ensure strict out-of-fold separation in each LOTO fold: (i) all crystal-pretrained network weights, descriptor definitions, physical signs, and equations (1) and (2) were fixed prior to benchmark evaluation; (ii) global cross-target feature medians and standard deviations (as well as inner-LOTO shrinkage weights for the `Reg. LOTO` sensitivity variant and ridge hybrid weights on Overath) were computed strictly from the 12 training targets without access to the held-out target; and (iii) on the held-out target t (cohort size N_t), only unlabeled candidate coordinates are used to compute within-target feature medians $\mathrm{med}_t(f_{\cdot,k})$ and standard deviations $\mathrm{sd}_t(f_{\cdot,k})$ via $z_{i,k}^{(t)} = \tanh((f_{i,k} - \mathrm{med}_t(f_{\cdot,k}))/(1.5\ \mathrm{sd}_t(f_{\cdot,k})))$ (whereas the $M = 1$ `True-EBM` variant uses zero test-cohort statistics). Cross-target and within-target subscores $s_i^{\mathrm{global}} = \mathbf{z}_i^{\mathrm{global}} \cdot \mathbf{w}_{\mathrm{global}}$ and $\hat{s}_i^{\mathrm{within}} = (s_i^{\mathrm{within}} - \mathrm{med}_t(s^{\mathrm{within}}))/\mathrm{sd}_t(s^{\mathrm{within}})$ are combined via:

$$S_i = \mathrm{med}_t\big(s^{\mathrm{global}}\big) + \beta_t\big(s_i^{\mathrm{global}} - \mathrm{med}_t(s^{\mathrm{global}})\big) + 0.75\ \mathrm{sd}_t\big(s^{\mathrm{global}}\big)\, \psi\big(\hat{s}_i^{\mathrm{within}}\big), \tag{1}$$

where $\beta_t = 0.40 \max(0, (N_t - 150)/(N_t + 400))$ and $\psi(u) = u + 0.20 \max(0, u - 0.8)^2$. For locked external post-cofold hybrids (`Nipah-G` and `Scorpion`), the locked Z-score $Z_{\text{BC-Score},i}$ modulates co-folder confidence $S_{\mathrm{cofold},i}$ via:

$$S_{\mathrm{hybrid},i} = S_{\mathrm{cofold},i} \cdot \exp\Big(0.25\ \tanh\big(Z_{\text{BC-Score},i}/1.5\big)\Big). \tag{2}$$

Ablating the constants in equations (1) and (2) on Overath LOTO predictions shows minimal sensitivity: setting $\psi(u) = u$ yields Pooled AP = 0.379 (13-target Mean AP = 0.560); setting both $\psi(u) = u$ and $\beta_t = 0$ yields 0.361 (0.555); and an uncalibrated affine sum $\mathrm{med}_t(s^{\mathrm{global}}) + \mathrm{sd}_t(s^{\mathrm{global}})\hat{s}_i^{\mathrm{within}}$ yields 0.369 (0.555). On `Nipah-G`, replacing equation (2) with $Z_{\mathrm{cofold}} + 0.25 Z_{\text{BC-Score}}$ yields AP = 0.217 and 14/50 top-50 hits.

4.4 Pipeline runtime profiling

Executed in batched GPU mode on a single NVIDIA A100-SXM4-40GB GPU (M = 25–32 complexes per disjoint-graph batch, matching the GPU hardware used for `Boltz-1` [35 s] and `AlphaFold3` [180 s]), amortized post-design `BC-Score` throughput on generated complex PDB structures (excluding upstream backbone generation, sequence design, and wet-lab testing) is: (1) **43.1** ms wall-clock per complex (23.2 complexes/s) on Overath ($\sim$ 200-aa targets; 41.4 ms with cached `ESM-2`). Across the five GPU compute passes, per-complex times are 11.73 ms for Stage A orientation and `ESM-2` (1.73 ms), 5.10 ms for Stage B monomer physics, 15.66 ms for Stage C sidechain MDN, 24-pose re-docking, and alchemical counterfactuals, 5.74 ms for disjoint autograd backbone forces, and 3.52 ms for continuous 6-DoF relaxation ($5.74 + 3.52 = 9.26$ ms for forces and relaxation; $11.73 + 5.10 + 15.66 + 9.26 = 41.75$ ms total GPU compute). Adding 1.35 ms for cached-receptor coordinate setup and host-to-device transfer yields the 43.10 ms (43.1 ms) batched wall-clock latency; if full uncached raw-PDB text parsing of both chains (2.70 ms) replaces the 1.35 ms cached-receptor setup, total per-complex time is $41.75 + 2.70 = 44.45$ ms (Fig. 5c). Relative to `AlphaFold3` (176.6 versus 0.042 A100 GPU-hours across 3,532 complexes) and `Boltz-1`, this corresponds to a **4,176**× and **812**× speedup at the post-design scoring step; (2) **37.6** ms batched wall-clock per complex on `Scorpion` ($\sim$ 65-aa targets; **4,783**× faster than `AlphaFold3` scoring); and (3) **41.4** ms batched wall-clock per complex on `Nipah-G` (532-aa receptor; **4,352**× faster than `AlphaFold3` scoring). For unbatched single-candidate ($M = 1$) evaluation on two CPU threads without a GPU, latency is 124.5 ms, 125.5 ms, and 201.8 ms per complex on Overath, `Scorpion`, and `Nipah-G` (7.3 min total CPU time for 3,532 complexes).

4.5 Statistical analysis

Ranking precision is measured by Average Precision ($\mathrm{AP} = \sum_k (R_k - R_{k-1}) P_k$) [19]. Confidence intervals (95%) and two-sided paired P values across the 3,532 Overath complexes were computed via $B = 10{,}000$ target-stratified paired bootstrap resamples (seed 2026) preserving per-target N_t and positive prevalence, with Benjamini–Hochberg FDR q-values [48] across all benchmark comparisons (Table 1 and Extended Data Table 2). Because target-stratified resampling conditions on the 13 observed campaigns, these intervals quantify candidate-level sampling variability within the benchmark rather than variance across future unseen target families. Same-epitope pairwise confidence intervals (Extended Data Fig. 2) used 1,000 binder-level cluster bootstraps; sidechain and co-design comparisons used two-sided Wilcoxon signed-rank and Mann–Whitney U tests.

4.6 Benchmark curation and reproducibility

Removing 7 unassigned rows and 137 multipartite peptide–MHC entries from the 3,676 rows in Overath et al. [19] yields all 3,532 single-chain receptor designs across 13 targets (391 binders, 11.07%). Subsampling unlabeled target batch sizes $M \in \{10, 25, 50, 100\}$ (25 replicates) confirmed stable 13-target Mean AP for `BC-Score` (0.536 ± 0.013 at $M = 10$, 0.552 ± 0.005 at $M = 25$, 0.554 ± 0.004 at $M = 50$, 0.555 ± 0.001 at $M = 100$, and 0.498 for within-target terms at $M = 1$ using training-target statistics). In strict $M = 1$ single-complex streaming mode without cohort statistics, a thermodynamic-cycle formulation (`True-EBM`, contrasting holo-versus-apo inverse folding, interface-versus-surface strain, and native-versus-decoy pose and alchemical partition functions under fixed training-set scales) reaches 0.409 13-target Mean AP on Overath (0.449 with `Boltz-1`, 0.585 with `AlphaFold3`), AP = 0.182 on `Nipah-G` (0.218 with `Boltz-2 ipSAE`), and 0.306 5-target Mean AP (AUROC = 0.651; 0.424 on publication hits) on `Scorpion` (versus 0.164 for unnormalized $M = 1$ `BC-Score`). Large language model tools (Google Gemini) were used to assist with code development, data analysis, and manuscript editing; all study design, model implementation, benchmarks, and statistical analyses were verified by the authors, who take full responsibility for the work.

Data availability. The Overath et al. benchmark dataset is publicly available from ref. [19], the `Nipah-G` campaign dataset is available from the Proteinbase *Nipah Competition Results* collection (`nipah-binder-competition-results`, https://proteinbase.com/collections/nipah-binder-competition-results), and the `Scorpion` neurotoxin binder design dataset is available on Hugging Face (`Anthropic/scorpion-toxin-binder-design`, https://huggingface.co/datasets/Anthropic/scorpion-toxin-binder-design, with target chains derived from PDB entries `25IH`, `1NPI`, `1AHO`, and `7K18` and AlphaFold Database entry `AF-P01479-F1`). All processed binding tables, precomputed structural feature caches, candidate-level out-of-fold predictions (n = 3,532 Overath complexes), locked external predictions (n = 881 `Nipah-G` complexes and n = 243 `Scorpion` complexes), and figure Source Data files are available in the anonymized repository at https://anonymous.4open.science/r/bc-score-2310/ and will be deposited in a permanent public repository upon publication. Source data are provided with this paper.

Code availability. All source code, pretrained PyTorch model checkpoints, training and evaluation scripts, and the SHA-256 frozen parameter manifest (`results/locked_v7_model_manifest.json`) are

available in the anonymized repository at https://anonymous.4open.science/r/bc-score-2310/ and will be released under an open-source license upon publication.

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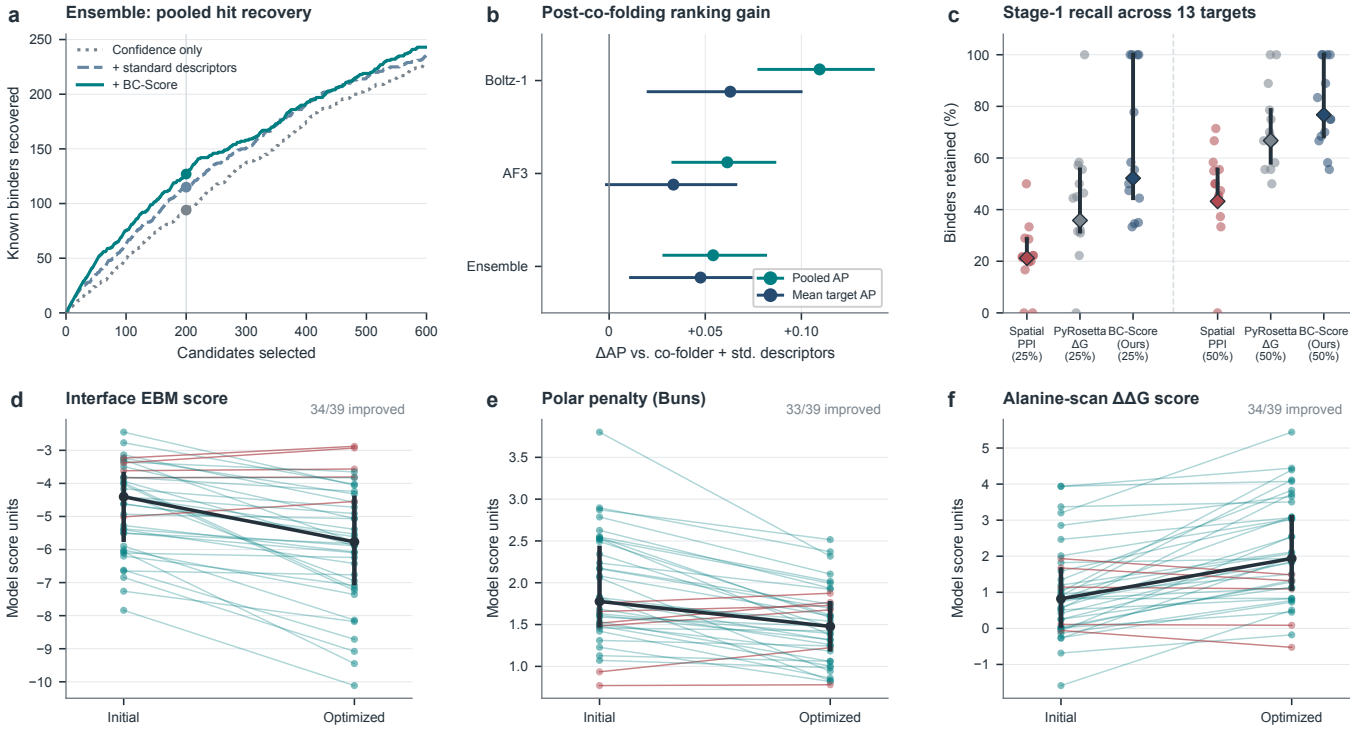
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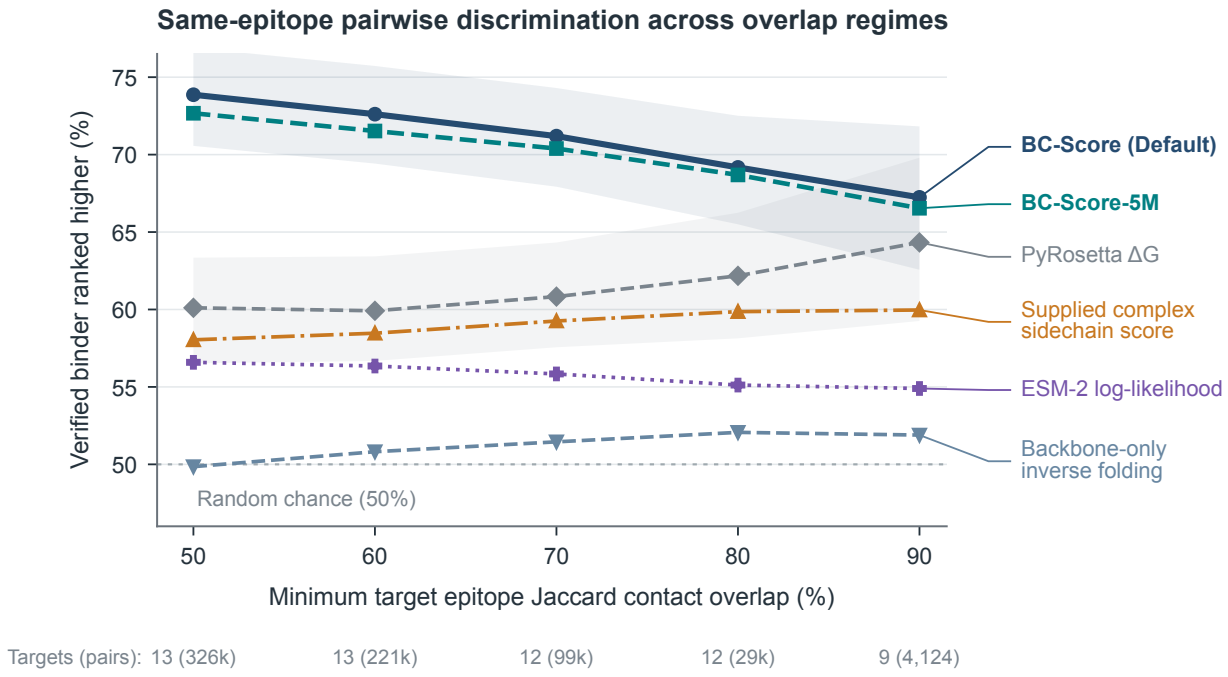
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Table 1. Summary of 13-Fold Leave-One-Target-Out (LOTO) Benchmark Across 3,532 Experimentally Tested De Novo Binders. Standalone evaluators operate strictly on input design PDB structures without co-folding (0.043 s per complex on an NVIDIA A100 GPU for `BC-Score` versus 35–180 s on the same A100 GPU for co-folding). Pooled Global AP is computed across all $n = 3,532$ complexes (391 verified binders); Mean Target AP is the unweighted mean across all 13 receptor campaigns; Multi-Hit Mean AP is the unweighted mean across the 11 campaigns with ≥ 6 binders ($n = 3,477$ complexes). Paired differences (ΔGlobal AP), two-sided paired bootstrap P values, and Benjamini–Hochberg FDR adjusted q-values across $B = 10,000$ target-stratified bootstrap resamples are reported relative to the indicated block reference. Extended Data Table 2 reports all 21 rows (including 95% confidence intervals and sensitivity variants).

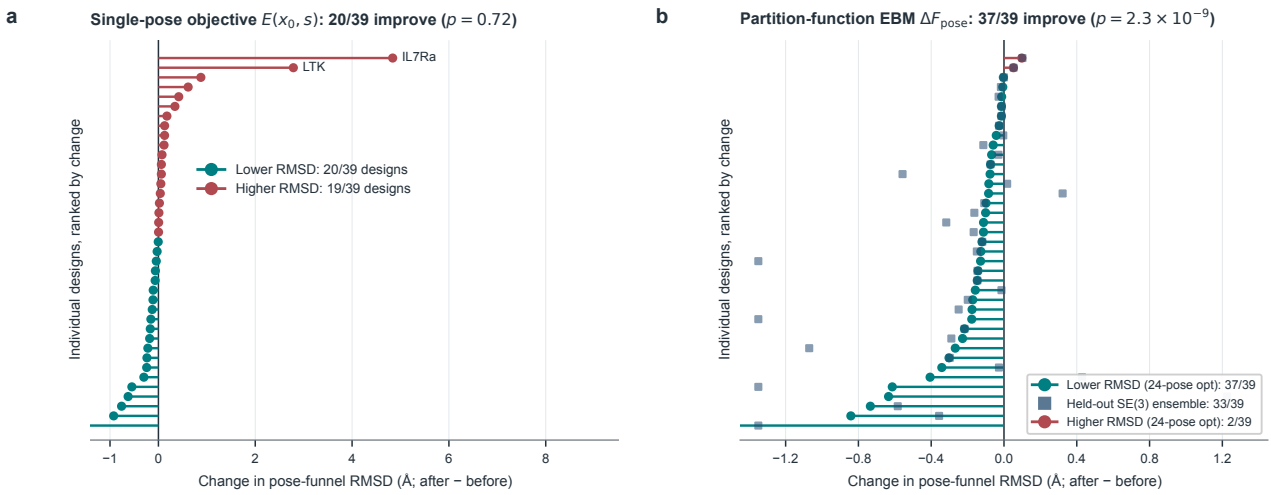
Method	Hardware and Latency	Pooled Global AP	Mean Target AP (All 13)	Multi-Hit Mean AP (11)	Paired ΔGlobal AP (P value; BH q)
Standalone Input-PDB Evaluators (Zero Co-Folding; versus PyRosetta Reference)					
`SpatialPPI v2` (3D GNN)	1.8 s CPU	0.120	0.174	0.180	-0.015 ($P = 0.146$; $q = 0.175$)
`PyRosetta` Interface ΔG_{sep}	4.5 s CPU	0.135	0.267	0.272	Reference (Input PDB)
`BC-Score-5M` (5-Module Equal-Weight)	0.043 s A100 GPU	0.366	0.527	0.483	$+0.230$ ($P < 0.001$; $q = 0.001$)
`BC-Score` **(Default, Equal-Weight)**	**0.043 s A100 GPU**	**0.383**	**0.559**	**0.502**	$\mathbf{+0.248}$ ($\mathbf{P < 0.001}$; $\mathbf{q = 0.001}$)
Standalone GPU Co-Folding Baselines (versus Standalone BC-Score Reference)					
`AlphaFold2-Initial-Guess iPAE`	GPU Co-Fold	0.242	0.398	0.341	-0.140 ($P < 0.001$; $q = 0.001$)
`ColabFold AF2-Multimer ipSAE`	GPU Co-Fold	0.289	0.376	0.412	-0.093 ($P < 0.001$; $q = 0.001$)
`Boltz-1 ipSAE`	35 s A100 GPU	0.336	0.425	0.430	-0.046 ($P = 0.036$; $q = 0.046$)
`AlphaFold3 ipSAE`	180 s A100 GPU	0.361	0.545	0.543	-0.022 ($P = 0.338$; $q = 0.375$)
Post-Co-Fold Hybrid Evaluators (versus Respective Raw Co-Folder Reference)					
`Boltz-1` + `BC-Score` **(Ours)**	**Boltz-1** + 0.043 s	**0.449**	**0.520**	**0.542**	$\mathbf{+0.113}$ ($\mathbf{P < 0.001}$; $\mathbf{q = 0.001}$)
`AlphaFold3` + `BC-Score` **(Ours)**	**AF3** + 0.043 s	**0.468**	**0.611**	**0.620**	$\mathbf{+0.107}$ ($\mathbf{P < 0.001}$; $\mathbf{q = 0.001}$)
`Boltz-1 + AF3` + `BC-Score` **(Ours)**	**Ensemble** + 0.043 s	**0.508**	**0.577**	**0.604**	$\mathbf{+0.136}$ ($\mathbf{P < 0.001}$; $\mathbf{q = 0.001}$)



Extended Data Figure 1. Supporting hybrid ranking, Stage-1 cascade recall, and internal optimization analyses. **a**, Pooled retrospective hit recovery across selection budgets for the `Boltz-1` and `AlphaFold3` ensemble ($n = 3,532$ complexes, 391 binders; grey dotted line marks random expectation), comparing confidence alone (grey dashed curve), standard Rosetta descriptors (blue curve), and `BC-Score` hybrid reranking (solid teal curve). At a budget of 200 candidates, 94, 115, and 127 binders are recovered, respectively. **b**, Point estimates (filled circles) and horizontal error bars showing 95% target-stratified paired bootstrap confidence intervals ($B = 10,000$ resamples across $n = 3,532$ complexes) for ΔAP (Pooled AP in teal, 13-target Mean AP in navy; two-sided paired bootstrap $P < 0.001$) when augmenting `Boltz-1`, `AlphaFold3`, or their ensemble with `BC-Score` instead of standard Overath Rosetta physicochemical descriptors. **c**, Per-target Stage-1 binder retention distributions across all 13 targets ($n = 391$ total binders) at retention budgets $\rho = 25\%$ (4.0× GPU speedup) and $\rho = 50\%$ (2.0× speedup), showing individual target recall values (jittered circles), interquartile ranges (vertical black lines), and pooled binder recall (filled diamonds) for `SpatialPPI v2` (red), `PyRosetta` ΔG_{sep} (grey), and `BC-Score` (navy). **d**–**f**, Paired changes under partition-function free-energy co-design in interface score (**d**, 34/39 improved, two-sided Wilcoxon signed-rank $P = 2.1 \times 10^{-8}$), buried unsatisfied polar penalty (**e**, 33/39 improved, $P = 8.9 \times 10^{-8}$), and alanine-scan specificity $\Delta\Delta G$ (**f**, 34/39 improved, $P = 2.1 \times 10^{-8}$) across $n = 39$ initially failed designs (3 per target). Filled teal circles denote improvement, filled red circles denote worsening, and black circles with vertical error bars show medians and interquartile ranges. Values reflect computational objective optimization on fixed backbones prior to experimental retesting.



Extended Data Figure 2. Fine-grained pairwise discrimination among same-epitope candidate binders. Point estimates and error bars (shaded bands and vertical bars) showing 95% cluster-bootstrap confidence intervals (1,000 resamples at the unique candidate binder level across $n = 3,532$ complexes) for pairwise ranking accuracy across all same-target binder and non-binder pairs sharing an increasingly identical receptor contact footprint ($C_\alpha < 8.5$ Å), from Jaccard contact overlap ≥ 0.50 ($n = 326,474$ pairs) to ≥ 0.90 ($n = 4,124$ pairs). Curves compare default `BC-Score` (solid teal curve with filled circles) and `BC-Score-5M` (dashed navy curve with open squares) against `PyRosetta` ΔG_{sep} (orange curve), supplied-sidechain scoring (red curve), `ESM-2` sequence log-likelihood (purple curve), and backbone-only inverse folding (grey curve; horizontal dashed line indicates 50% random chance; two-sided cluster-bootstrap $P < 0.001$ for `BC-Score` versus all baselines at overlap ≥ 0.50).



Extended Data Figure 3. Partition-function free-energy co-design resolves the single-pose funnel degradation mode across 39 failed designs. **a**, Paired change in pose-funnel RMSD (after minus before optimization) across all $n = 39$ initially failed Overath designs (3 per target) under single-pose coordinate descent ($E(\mathbf{x}_0, \mathbf{s})$), where 20 designs sharpen the pose funnel (filled teal bars, $\Delta\text{RMSD} < 0$) and 19 worsen (filled red bars, $\Delta\text{RMSD} > 0$; median $\Delta\text{RMSD} = -0.01$ Å, two-sided Wilcoxon signed-rank $P = 0.72$) due to unpenalized competing decoy basins. **b**, Paired change in pose-funnel RMSD across the same $n = 39$ designs under statistical-mechanics partition-function free-energy co-design ($\Delta F_{\text{pose}}(\mathbf{x}_0, \mathbf{s}) = E_\theta(\mathbf{x}_0, \mathbf{s}) - F_{\text{decoy}}(\mathbf{s})$), which sharpens the 24-pose funnel on **37** of **39** designs (filled teal bars; mean 0.95 Å → 0.74 Å, two-sided Wilcoxon signed-rank $P = 2.3 \times 10^{-9}$) and generalizes to an independent held-out 24-pose SE(3) verification ensemble on **33** of **39** designs (mean 2.22 Å → 1.06 Å, $P = 1.3 \times 10^{-4}$).

Extended Data Table 1. Held-Out Crystal Sidechain Reconstruction Accuracy ($n = 60$ Validation Monomers, $12,248$ Residues; $n = 50$ Crystal Complexes). Our 2-Point SE(3) GNN reconstructs sidechain centroid ($\hat{\mathbf{s}}_i^{\text{recon}}$) and distal functional tip ($\hat{\mathbf{t}}_i^{\text{recon}}$) coordinates strictly from backbone atoms ($\text{N}, \text{C}_\alpha, \text{C}, \text{O}, \text{C}_\beta$) and amino acid identity, outperforming a canonical rigid-backbone local-frame template across all residue classes and at crystal complex interfaces (< 8.5 Å C_α distance). Values report Mean Absolute Error (MAE, in Å) and the percentage of residues reconstructed within 1.0 Å of native crystal coordinates ($P < 10^{-15}$ by two-sided paired Wilcoxon signed-rank test across $n = 12,248$ residues).

	Canonical Rigid-Backbone Template			Pretrained 2-Point SE(3) GNN (Ours)		
Residue Subset	**Centroid MAE**	**Tip MAE**	**Within 1.0 Å**	**Centroid MAE**	**Tip MAE**	**Within 1.0 Å**
All Non-Gly and Non-Ala Monomer Residues (12,248)	0.89 Å	1.59 Å	65.6%	**0.48 Å**	**0.86 Å**	**86.0%**
Charged Surface and Interface (Arg, Lys, Asp, Glu)	1.08 Å	2.10 Å	51.4%	**0.68 Å**	**1.35 Å**	**76.1%**
Polar H-Bonding (Ser, Thr, Asn, Gln, His)	0.76 Å	1.41 Å	72.6%	**0.43 Å**	**0.93 Å**	**92.1%**
Bulky Aromatics (Phe, Trp, Tyr)	2.13 Å	3.81 Å	16.9%	**0.58 Å**	**0.90 Å**	**80.6%**
Crystal Complex Interface Residues (< 8.5 Å C_α)	1.05 Å	—	57.4%	**0.57 Å**	—	**81.3%**

Extended Data Table 2. Complete 13-Fold Leave-One-Target-Out (LOTO) Benchmark and Sensitivity Analysis Across $n = 3,532$ De Novo Binders. Standalone input-PDB evaluators operate strictly on input design PDB structures without co-folding (0.043 s per complex on an NVIDIA A100 GPU for `BC-Score` versus 35–180 s on the same A100 GPU for co-folding). Columns report Pooled Global AP ($n = 3,532$), unweighted 13-target Mean Target AP, 11-target Multi-Hit Mean AP ($n = 3,477$), and Pooled AP excluding `FGFR2` ($n = 1,409$). Paired differences (ΔGlobal AP), 95% confidence intervals, two-sided paired bootstrap P values ($B = 10,000$ target-stratified resamples), and Benjamini–Hochberg FDR q-values are reported relative to the reference baseline indicated in each block.

Method or Variant	**Fitted Weights**	**Pooled Global AP**	**Mean Target AP (All 13)**	**Multi-Hit AP (11)**	**Pooled AP (Excl. FGFR2)**	**Paired ΔGlobal AP [95% CI] (P value; BH q)**
Standalone Input-PDB Evaluators (Zero Co-Folding; versus PyRosetta Reference)						
`SpatialPPI v2` (3D GNN) [23]	Pretrained	0.120	0.174	0.180	0.165	−0.015 [−0.036, +0.005] ($P = 0.146$; $q = 0.175$)
`PyRosetta` Interface ΔG_{sep} [21]	0	0.135	0.267	0.272	0.198	Reference (Input PDB)
`True-EBM` ($M = 1$ Cohort-Free Streaming)	0	0.233	0.409	0.366	0.312	+0.098 [+0.068, +0.129] ($P < 0.001$; $q = 0.001$)
`BC-Score-5M` (5-Module, Equal-Weight)	0	0.366	0.527	0.483	0.479	+0.230 [+0.200, +0.261] ($P < 0.001$; $q = 0.001$)
`BC-Score-5M` (5-Module, Nested LOTO)	5 (Inner)	0.364	0.537	0.480	0.478	+0.229 [+0.198, +0.259] ($P < 0.001$; $q = 0.001$)
`BC-Score` **(Default, Equal-Weight)**	**0**	**0.383**	**0.559**	**0.502**	**0.487**	+0.248 [+0.217, +0.280] **($P < 0.001$; $q = 0.001$)**
`BC-Score` **(Regularized LOTO)**	**Shrinkage**	**0.382**	**0.560**	**0.503**	**0.484**	+0.247 [+0.218, +0.279] **($P < 0.001$; $q = 0.001$)**
Standalone GPU Co-Folding Baselines (versus Standalone BC-Score Default Reference)						
`AlphaFold2-Initial-Guess iPAE` [18]	0	0.242	0.398	0.341	0.242	−0.140 [−0.172, −0.109] ($P < 0.001$; $q = 0.001$)
`ColabFold AF2-Multimer ipSAE` [9, 38]	0	0.289	0.376	0.412	0.327	−0.093 [−0.129, −0.057] ($P < 0.001$; $q = 0.001$)
`ColabFold AF2-Multimer actifpTM` [39]	0	0.295	0.380	0.425	0.357	−0.087 [−0.128, −0.046] ($P < 0.001$; $q = 0.001$)
`Boltz-1 ipSAE` [11, 38]	0	0.336	0.425	0.430	0.353	−0.046 [−0.086, −0.004] ($P = 0.036$; $q = 0.046$)
`AlphaFold3 ipTM` [10]	0	0.319	0.480	0.466	0.346	−0.063 [−0.102, −0.024] ($P = 0.004$; $q = 0.006$)
`AlphaFold3 LIS` [40]	0	0.350	0.506	0.496	0.362	−0.032 [−0.077, +0.009] ($P = 0.126$; $q = 0.162$)
`AlphaFold3 ipSAE` [10, 38]	0	0.361	0.545	0.543	0.390	−0.022 [−0.064, +0.023] ($P = 0.338$; $q = 0.375$)
Post-Co-Fold Hybrid Evaluators (versus Respective Raw Co-Folder Reference)						
`Boltz-1` + Std. `Rosetta` Phys. [19]	LOTO	0.339	0.457	0.467	0.330	+0.003 [−0.019, +0.025] ($P = 0.780$; $q = 0.780$)
`Boltz-1` + `BC-Score` **(Ours)**	**LOTO**	**0.449**	**0.520**	**0.542**	**0.474**	+0.113 [+0.078, +0.145] **($P < 0.001$; $q = 0.001$)**
`AlphaFold3` + Std. `Rosetta` Phys. [19]	LOTO	0.407	0.577	0.581	0.439	+0.046 [+0.019, +0.074] ($P < 0.001$; $q = 0.001$)
`AlphaFold3` + `BC-Score` **(Ours)**	**LOTO**	**0.468**	**0.611**	**0.620**	**0.537**	+0.107 [+0.072, +0.138] **($P < 0.001$; $q = 0.001$)**
`Boltz-1 + AF3` (Raw Ensemble)	0	0.372	0.523	0.541	0.396	Reference (Ensemble)
`Boltz-1 + AF3` + Std. `Rosetta` Phys.	LOTO	0.454	0.530	0.549	0.450	+0.082 [+0.051, +0.112] ($P < 0.001$; $q = 0.001$)
`Boltz-1 + AF3 + BC-Score` **(Ours)**	**LOTO**	**0.508**	**0.577**	**0.604**	**0.550**	+0.136 [+0.103, +0.169] **($P < 0.001$; $q = 0.001$)**

Extended Data Table 3. Retrospective Two-Stage Screening Cascade Simulation Across $n = 3,532$ Candidates and 13 Targets (391 Verified Binders). For each Stage-1 filter and per-target retention fraction ρ, candidates are ranked within each target by their out-of-fold Stage-1 score, and only the top $\lceil \rho N_t \rceil$ survivors are passed to Stage-2 `AlphaFold3` co-folding (180 s per complex on an NVIDIA A100 GPU). Columns report Stage-1 pooled binder recall, across-target mean and median recall ($n = 13$ targets), and verified binders recovered in the Top-10 per target (maximum 130) and Global Top-200 (in parentheses). Per-target recall distributions are shown in Extended Data Fig. 1c.

Stage-1 Filter	Co-Fold Budget ρ (n)	Total Compute (GPU Reduction)	Stage-1 Pooled Recall	Target Recall (Mean, Median)	AF3 Alone Top-10/T (Top-200)	AF3 + BC-Score Top-10/T (Top-200)
Reference Workflow: Co-Fold 100% of Candidates (No Pre-Filter)						
None (100% Co-Folded)	100% (3,532)	176.6 h GPU (1.0×)	100.0%	100.0%, 100.0%	67 (89)	**74 (123)**
Zero Co-Folding Workflows (Stage-1 Filter Alone; 0% Co-Folded)						
`SpatialPPI v2` Alone	0% (0)	1.8 h CPU	—	—	20 (32)	—
`PyRosetta` ΔG_{sep} Alone	0% (0)	4.4 h CPU	—	—	33 (25)	—
`BC-Score` **(Default, Equal) Alone**	**0% (0)**	**0.042 h GPU** (4,176×)	—	—	**56 (111)**	—
`BC-Score` **(Reg. LOTO) Alone**	**0% (0)**	**0.042 h GPU** (4,176×)	—	—	**58 (110)**	—
Two-Stage Screening Cascade at $\rho = 25\%$ Co-Folding Budget (4.0× Fewer GPU Calls)						
`SpatialPPI v2` → Co-Fold	25% (888)	44.4 h GPU (4.0×)	21.2%	21.9%, 21.8%	35 (52)	29 (55)
`PyRosetta` ΔG_{sep} → Co-Fold	25% (888)	44.4 h GPU (4.0×)	35.8%	44.1%, 45.0%	55 (92)	59 (88)
`BC-Score` **(Default) → Co-Fold**	**25% (888)**	**44.4 h GPU** (4.0×)	**52.2%**	**64.3%, 55.4%**	**66 (113)**	**69 (119)**
`BC-Score` **(Reg. LOTO) → Co-Fold**	**25% (888)**	**44.4 h GPU** (4.0×)	**52.7%**	**64.6%, 56.0%**	**67 (116)**	**69 (119)**
Two-Stage Screening Cascade at $\rho = 40\%$ and $\rho = 50\%$ Co-Folding Budgets (2.5× and 2.0× Fewer GPU Calls)						
`SpatialPPI v2` → Co-Fold	50% (1,770)	88.5 h GPU (2.0×)	43.2%	47.3%, 50.0%	54 (85)	54 (79)
`PyRosetta` ΔG_{sep} → Co-Fold	50% (1,770)	88.5 h GPU (2.0×)	66.8%	71.8%, 68.4%	64 (102)	70 (111)
`BC-Score` **(Default) → Co-Fold**	**40% (1,419)**	**71.0 h GPU** (2.5×)	**65.5%**	**73.3%, 71.0%**	**71 (107)**	**71 (120)**
`BC-Score` **(Default) → Co-Fold**	**50% (1,770)**	**88.5 h GPU** (2.0×)	**76.7%**	**80.1%, 75.0%**	**75 (107)**	**72 (122)**
`BC-Score` **(Reg. LOTO) → Co-Fold**	**50% (1,770)**	**88.5 h GPU** (2.0×)	**76.7%**	**80.0%, 75.0%**	**75 (107)**	**72 (122)**

Extended Data Table 4. Held-Out External Validation on Two Public Experimental Campaigns. All `BC-Score` feature definitions, signs, checkpoints, and Overath-fitted normalization parameters were frozen prior to evaluation and applied without modification. On `Nipah-G` (`nipah-binder-competition-results`; $n = 881$ `Boltz-2`-predicted complexes [12], 100 binders), Pooled AP is shown with 95% bootstrap confidence intervals ($B = 10,000$ resamples) in brackets, alongside AUROC and top-K hit recovery at $K = 25$ and $K = 50$. On `Scorpion` (`Anthropic/scorpion-toxin-binder-design`; $n = 243$ raw upstream pre-co-fold design PDB structures across 5 targets; `ESMFold2` denotes the CZ Biohub `ESMFold2` complex predictor [45]), point estimates are reported across all 21 Adaptyv Bio binding hits with sensitivity point estimates on the 12 Twist-confirmed publication hits shown in brackets, and top-K recovery is reported at $K = 20$ and $K = 50$.

Campaign and Method	Input and Category	Pooled AP [Nipah 95% CI / Scorpion 12-Hit Pub]	Mean Target AP [Scorpion 12-Hit Pub]	AUROC	Top-25 or Top-20 Hits	Top-50 Hits
External Campaign 1: Adaptyv Bio Nipah-G *($n = 881$* ***Boltz-2 Complexes,*** *100* ***Binders [11.35%], 1 Target; Brackets = 95% Bootstrap CI)***						
`ProteinMPNN` Sequence Score [4]	Sequence Log-Prob	0.100 [0.083, 0.124]	0.100	0.479	1 / 25 (4%)	2 / 50 (4%)
`Boltz-2 pDockQ2` [44]	GPU Co-Folder Metric	0.114 [0.093, 0.134]	0.114	0.500	4 / 25 (16%)	4 / 50 (8%)
`ESMFold` Monomer pLDDT [42]	Fast Monomer Fold	0.124 [0.097, 0.170]	0.124	0.548	2 / 25 (8%)	3 / 50 (6%)
`Boltz-2 LIS` [40]	GPU Co-Folder Metric	0.131 [0.105, 0.167]	0.131	0.583	0 / 25 (0%)	0 / 50 (0%)
`Boltz-2 ipTM` [12]	GPU Co-Folder Metric	0.136 [0.108, 0.178]	0.136	0.587	2 / 25 (8%)	4 / 50 (8%)
`Boltz-2 Complex pLDDT` [12]	GPU Co-Folder Metric	0.146 [0.115, 0.202]	0.146	0.594	5 / 25 (20%)	7 / 50 (14%)
`BC-Score` **(Default, Equal) [Locked]**	**Boltz-2 Struct** (0.041 s)	**0.155 [0.116, 0.213]**	**0.155**	**0.567**	**3 / 25 (12%)**	**10 / 50 (20%)**
`BC-Score` **(Reg. LOTO) [Locked]**	**Boltz-2 Struct** (0.041 s)	**0.156 [0.118, 0.217]**	**0.156**	**0.563**	**4 / 25 (16%)**	**11 / 50 (22%)**
`True-EBM` ($M = 1$ Streaming)	Boltz-2 Struct (0.041 s)	0.182 [0.135, 0.244]	0.182	0.594	5 / 25 (20%)	11 / 50 (22%)
`Boltz-2 ipSAE` [12, 38]	GPU Co-Folder Metric	0.193 [0.142, 0.262]	0.193	0.622	7 / 25 (28%)	9 / 50 (18%)
`Boltz-2` min ipSAE [38]	GPU Co-Folder Metric	**0.229 [0.164, 0.299]**	**0.229**	**0.645**	**9 / 25 (36%)**	10 / 50 (20%)
Hybrid: `Boltz-2 ipSAE + BC-Score`	**Post-Co-Fold Rerank**	**0.229 [0.162, 0.311]**	**0.229**	**0.618**	**8 / 25 (32%)**	**16 / 50 (32%)**
External Campaign 2: Anthropic and Adaptyv Scorpion Neurotoxins *($n = 243$* ***Raw PDBs, 5 Targets; 21 Adaptyv [12 Pub] Hits)***						
`Protenix-V2 ipSAE` (AF3 Reprod.) [13]	GPU Co-Folder	0.226 [0.174]	0.262 [0.224]	0.748	5 / 20 [3/20]	11 / 50 [7/50]
`Protenix-V2 scDockQ` [13, 46]	GPU Co-Folder	0.222 [0.145]	0.305 [0.211]	0.805	2 / 20 [1/20]	14 / 50 [9/50]
`ESMFold2 scDockQ` [45, 46]	Fast Complex Fold	0.231 [0.153]	0.290 [0.219]	0.819	5 / 20 [3/20]	13 / 50 [7/50]
Campaign Multi-Oracle Composite	Multi-Oracle Pipeline	0.264 [0.186]	0.329 [0.296]	**0.824**	5 / 20 [2/20]	14 / 50 [10/50]
`ESMFold2 ipSAE` [45, 38]	Fast Complex Fold	0.336 [0.289]	0.377 [0.386]	0.823	8 / 20 [6/20]	16 / 50 [11/50]
`BC-Score` **(Default, Equal) [Locked]**	**Raw PDB** (0.038 s)	**0.141 [0.104]**	**0.319 [0.448]**	**0.618**	**3 / 20 [2/20]**	**7 / 50 [4/50]**
`BC-Score-5M` (5-Module, Equal) [Locked]	Raw PDB (0.038 s)	0.170 [0.165]	0.408 [0.599]	0.609	3 / 20 [2/20]	7 / 50 [5/50]
`True-EBM` ($M = 1$ Streaming)	Raw PDB (0.038 s)	0.152 [0.123]	0.306 [0.424]	0.651	3 / 20 [3/20]	8 / 50 [5/50]
Hybrid: `Protenix-V2 + BC-Score`	**Post-Co-Fold Rerank**	**0.288 [0.275]**	**0.367 [0.415]**	**0.706**	**8 / 20 [6/20]**	**11 / 50 [8/50]**
Hybrid: `ESMFold2 + BC-Score`	**Post-Co-Fold Rerank**	**0.376 [0.365]**	**0.463 [0.527]**	**0.785**	**10 / 20 [8/20]**	**13 / 50 [9/50]**